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Seungwook Kim

Research Scientist, PhD

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I am Seungwook Kim, a computer vision researcher and Tech Lead at [GenGenAI](#), where I work on generating data with AI, for AI. My work has centered on understanding and generating visual structure: from visual correspondence in 2D images and 3D point clouds, to 2D/3D equivariance, to generative models for images, video, and 3D. I am currently interested in any-modality-to-any-modality generation, world models, and self-improving AI systems that use intrinsic signals to improve their own fidelity and reliability. I conducted my doctoral research at the POSTECH Computer Vision Lab under Prof. [Minsu Cho](#).

TECHNICAL EXPERIENCE

Tech Lead - Alternative Military Service (Special Research Agent) **04 2026 — Present**
GenGenAI *Seoul, South Korea*

- Built scalable pipelines for generating synthetic image and video datasets across EO/IR modalities in data-scarce scenarios.
- Developed self-evaluation methods that use intrinsic signals from generative models for automated quality assessment.
- Developing a foundation model for scalable and controllable synthetic data generation across scenarios and modalities.

PhD Intern / Seed-Vision-Image Generation Team **06 2024 — 12 2024**
ByteDance *CA, USA*

- Developed methods to improve cross-view consistency and inference efficiency in text,image-to-multiview generation models.
- Published related work, RapidMV, at WACV 2026.

PhD Intern / Data-Intelligent Creation-Vision and Graphics Team **09 2023 — 02 2024**
ByteDance *CA, USA*

- Worked on text,image-to-3D generation models, with a focus on improving geometric consistency and controllability.
- Published related work in CorrespondentDream (CVPR 2024) and MultiImageDream (arXiv 2024).

Undergraduate Intern / 3D Map Construction from LiDAR **03 2020 — 07 2020**
Polaris3D *Pohang, South Korea*

- Built a real-time data acquisition pipeline from Intel RealSense cameras to NVIDIA Jetson Nano.
- Fused dual-view sensor streams into a real-time 3D mapping pipeline.

Undergraduate Intern / Camera ISP **12 2019 — 01 2020**
SK Hynix *Icheon, South Korea*

- Analyzed image signal processing algorithms applied to raw sensor images.
- Investigated vignetting-related intensity imbalance in raw images from a prototype sensor.

Undergraduate Intern / AI Team **06 2019 — 08 2019**
Netmarble *Seoul, South Korea*

- Optimized a speech-to-3D lip synthesis pipeline for lightweight mobile deployment using TensorFlow.

Undergraduate Intern / Data Engineering & Analysis Team **02 2018 — 12 2018**
Dable *Seoul, South Korea*

- Analyzed high-volume raw traffic data in AWS Redshift using PostgreSQL.
- Implemented scheduled batch jobs to process Redshift data and export results to a MySQL server.

EDUCATION

PhD Candidate / Research Scientist **08 2020 — 02 2026**
POSTECH Computer Vision Lab *Pohang, South Korea*

- {Image/Text}-to-{3D/Video} Generation (joint work with [Bytedance Seed](#))
- 2D Visual Correspondence, a.k.a wide-baseline/semantic image matching.
- 3D Visual Correspondence, a.k.a point cloud registration / geometric assembly
- 2D / 3D equivariance to rotation / scale (joint work with [Samsung MX](#))

Bachelor of Engineering in Computer Sciences and Engineering, POSTECH **07 2020**
Cumulative GPA: 3.7 / 4.3

Jigok Scholarship (Full scholarship) **2015 — 2020**

Student Mentoring Program Scholarship (Monthly scholarship) **2017 — 2020**

On-school Work Scholarship (Per-semester scholarship) **2017 — 2020**

Bachelor of Engineering in Computer Sciences and Engineering, Seoul National University **12 2017 — 01 2018**
Winter session

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PUBLICATIONS

- [C11] Improving Text-to-Image Generation with Intrinsic Self-Confidence Rewards Synthesis CVPR 2026
Seungwook Kim, Minsu Cho
- [W4] FreeAction: Training-Free Techniques for Enhanced Fidelity of Trajectory-to-Video Generation CoRL LSRW Workshop 2025
Seungwook Kim, Seunghyeon Lee, Minsu Cho
- [C10] RapidMV: Leveraging Spatio-Angular Representations for Efficient and Consistent Text-to-Multi-View Synthesis WACV 2026
Seungwook Kim, Yichun Shi, Kejie Li, Minsu Cho, Peng Wang
- [W3] Similarity-Aware Selective State-Space Modeling for Semantic Correspondence ICCV Findings 2025
Seungwook Kim, Minsu Cho **Oral Presentation**
- [W2] Harnessing the Power of Training-Free Techniques for Text-to-3D Generation via Score Distillation Sampling CVPR AI4CC 2025
Junhong Lee, Seungwook Kim, Minsu Cho
- [C9] 3D Geometric Shape Assembly via Efficient Point Cloud Matching ICML 2024
Nahyuk Lee, Juhong Min*, Junha Lee, Seungwook Kim, Kanghee Lee, Jaesik Park, Minsu Cho*
- [P1] Multi-view Image Prompted Multi-view Diffusion for Improved 3D Generation Arxiv 2024
Seungwook Kim, Yichun Shi, Kejie Li, Minsu Cho, Peng Wang
- [C8] Enhancing 3D Fidelity of Text-to-3D using Cross-View Correspondences CVPR 2024
Seungwook Kim, Kejie Li, Xueqing Deng, Yichun Shi, Minsu Cho, Peng Wang
- [C7] Learning SO(3)-Invariant Semantic Correspondence via Local Shape Transform CVPR 2024
Chunghyun Park, Seungwook Kim*, Jaesik Park, Minsu Cho*
- [C6] Efficient Semantic Matching with Hypercolumn Correlation WACV 2024
Seungwook Kim, Juhong Min, Minsu Cho **Best paper finalist**
- [C5] Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning ICML 2023
Seungwook Kim, Chunghyun Park*, Yoonwoo Jeong, Jaesik Park, Minsu Cho*
- [C4] Learning Rotation-Equivariant Features for Visual Correspondence CVPR 2023
Jongmin Lee, Byungjin Kim, Seungwook Kim, Minsu Cho
- [J1] Convolutional Hough Matching Networks for Robust and Efficient Visual Correspondence TPAMI 2023
Juhong Min, Seungwook Kim, Minsu Cho
- [W1] SeLCA: Self-Supervised Learning of Canonical Axis NeurReps Workshop 2022
Seungwook Kim, Yoonwoo Jeong, Chunghyun Park, Jaesik Park, Minsu Cho
- [C3] TransforMatcher: Match-to-Match Attention for Semantic Correspondence CVPR 2022
Seungwook Kim, Juhong Min, Minsu Cho
- [C2] Deep Hough Voting for Robust Global Registration ICCV 2021
Junha Lee, Seungwook Kim, Minsu Cho, Jaesik Park
- [C1] Learning to Distill Convolutional Features into Compact Local Descriptors WACV 2021
Jongmin Lee, Yoonwoo Jeong, Seungwook Kim, Juhong Min, Minsu Cho

INVITED TALKS / MENTORING

- Invited talk, Chungang University (Hosted by Prof. Jongmin Lee)** 03 2026
Delivered talk on my PhD Thesis: **Improving the Fidelity of Generative Diffusion Models using Unsupervised Intrinsic Guidance**, spanning CorrespondentDream (CVPR 2024), RapidMV (WACV 2026), FreeAction (CoRL 2025), and SOLACE (CVPR 2026).
- Research outline talk, Korean Electronics Technology Institute** 04 2023
Delivered talk on TransforMatcher: Match-to-Match Attention for Semantic Correspondence (CVPR 2022) / Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning (ICML 2023)
- Identifying Match-wise relationships for Semantic Correspondence, Samsung Advanced Institute of Technology** 01 2023
Delivered talk on TransforMatcher: Match-to-Match Attention for Semantic Correspondence (CVPR 2022)
- Identifying Match-wise relationships for Semantic Correspondence, Korean Conference on Computer Vision** 08 2022
Delivered talk on TransforMatcher: Match-to-Match Attention for Semantic Correspondence (CVPR 2022)
- Learning Canonical Axis of 3D Objects in a Self-Supervised Manner, Samsung Advanced Institute of Technology** 06 2022
Delivered talk on SeLCA: Self Supervised Learning of Canonical Axis (NeurReps Workshop 2022)
- POSTECH Tech Review, POSTECH** 06 2021 / 06 2022
Delivered talk on adversarial examples and AI safety (2021), and 2D image correspondences (2022)
- CORE CSE Undergraduate Mentoring, POSTECH** 03 2024 — 05 2024
Mentored 3 undergraduate CSE students on various aspects (study, research, life plans)

HONOURS & AWARDS

Outstanding Reviewer, CVPR Acknowledged as an Outstanding Reviewer for CVPR 2026	06 2026
Outstanding Reviewer, ICCV Acknowledged as an Outstanding Reviewer for ICCV 2025	03 2025
The POSTECHIAN Fellowship - Innovation, POSTECH Additional 6000 USD award in cash	12 2024
Best Paper Award, IPIU Best Paper Award @ IPIU 2024, South Korea (1/311 submitted papers)	02 2024
Best Paper Candidate, WACV Best Paper Candidate @ WACV 2024, Big Island (Top 0.6%)	01 2024
Hyundai Motor Chung Mong-Koo Foundation Scholarship, Hyundai Motor Approximately 13,000 USD per year	07 2023 — 08 2025
BK21 Outstanding Research Paper Awards, POSTECH Additional 500 USD award in cash	01 2023
The POSTECHIAN Fellowship - Leadership, POSTECH Additional 1000 USD award in cash	12 2022
BK21 Outstanding Research Paper Awards, POSTECH Additional 500 USD award in cash	01 2022
NAVER AI RUSH: 1st runner up in click-through rate (CTR) prediction, NAVER Additional 7,000 USD award in cash	05 2020 — 07 2020
POSTECH Creative Self-Research Scholarship, POSTECH 5,000 USD funding for self-research	09 2020 — 12 2020
NAVER AI RUSH: 1st runner up in image classification with noisy labels, NAVER Additional 4,000 USD award in cash	07 2020 — 08 2020

ACADEMIC ACTIVITIES

Journal / Conference Reviewer • CVPR (2022-2026), ECCV(2022,2024), ICCV (2023, 2025), NeurIPS (2023-2025), ICML (2024-2025) • 3DV (2022), WACV (2022-2026), ACCV (2024)	08 2020 — Present
Technical Chair ACCV 2024 • Managed Microsoft CMT for coordinating the paper submission and review process.	05 2024 — 12 2024 Hanoi, Vietnam

PATENTS

- **Methods and Systems for Generating a Multi-Dimensional Image Using Cross-View Correspondences.** US Patent Application Publication No. US20250239005A1; US Application No. 18/418,215; PCT Application No. PCT/SG2024/050823.
- **Apparatus and Method with Image Matching.** US Application No. 18/316,347; KR Application No. 10-2022-0150904.
- **Prediction Method for Characteristic Orientation of Point Clouds and Analysis Apparatus.** US Application No. 18/967,459; KR Application No. 10-2023-0176523.

REFERENCES

Donghoon Yeo CTO, GenGenAI	paul@gengen.ai 2026.04 — Present
Kejie Li Intern Manager / Host, ByteDance Research	liko.jack@gmail.com 2024.06 — 2024.12
Peng Wang Intern Manager / Host, ByteDance Research	pengwangpku2012@gmail.com 2023.09 — 2024.02
Minsu Cho Ph.D. Advisor, POSTECH	mscho@postech.ac.kr 2020.09 — 2026.03